District Flood Validation by Rank-Based Fusion of Sentinel-1 and Humanitarian Social Signals

Abstract*—Satellite flood maps and humanitarian social media content describe different aspects of a disaster. This study tests whether these two sources can be combined into a transparent district priority list without treating either one as ground truth. For the 2018 Kerala flood, Sentinel-1 VV backscatter change was mapped across 81 processing tiles using sensitive (-2.0 dB), balanced (-2.5 dB), and conservative (-3.0 dB) thresholds. HumAID records were deduplicated using exact text matching rather than semantic similarity, leaving 7,984 records. Of these, 1,080 (13.53%) could be assigned to a Kerala district, and only 305 records with impact or need labels were retained for scoring, representing 3.82% of the deduplicated dataset. This low retention rate is a key operational constraint arising from limited district resolution and label filtering. District flood area and retained record counts were converted to within-event ranks before weighted fusion. Statewide mapped area varied from 219.42 to 344.74 km², while district ordering remained stable, with pairwise Spearman correlations of 0.982 or higher. The balanced SAR ranking and the humanitarian ranking showed only a weak association ($\rho = 0.279$, permutation $p = 0.330$), indicating that the two sources emphasize different districts. At equal weight, Alappuzha, Thrissur, Ernakulam, Kottayam, and Wayanad ranked highest. The result is an auditable screening order rather than a validated allocation of relief.

***Index Terms*—Sentinel-1 SAR, flood mapping, social sensing, HumAID, rank aggregation, disaster response.**

I. INTRODUCTION

Flood response teams must compare evidence sources that carry different meanings. Sentinel-1 images measure changes in radar backscatter that can indicate surface inundation at the time of acquisition [1]-[3]. Humanitarian social media content can describe evacuation, infrastructure damage, injury, or urgent requests, information that is difficult to infer from satellite images alone [4], [5]. Neither source provides a complete account. SAR may miss water under vegetation, in built-up areas, or between acquisitions. Online reports appear only where people post and where a place name can be recovered [6]. In dense urban districts such as Ernakulam, double-bounce scattering and radar shadow can mask or mimic flood-related backscatter changes, producing false negatives or false positives and weakening correspondence with social media reports.

The 2018 Kerala flood provides a relevant test case. Cloud cover limits optical mapping during monsoon conditions, while Sentinel-1 supplies repeat C-band observations [7], [9]. Humanitarian records add information about reported consequences. This study therefore asks a narrower question than whether the two sources agree. Can their district-level evidence be combined in a way that preserves their differences and makes the effect of weighting visible?

The two sources are processed independently and joined only after district aggregation. Satellite evidence is expressed as mapped area under three change thresholds. Humanitarian evidence is expressed as the count of district-resolved HumAID records with impact or need labels. Rank normalization avoids adding square kilometres directly to record counts. Because online participation and geolocation are uneven, the social measure is interpreted as observed reporting activity rather than population-level severity [6].

II. RELATED WORK

A. Flood Mapping with SAR

Sentinel-1 GRD supports rapid flood mapping because it provides calibrated backscatter under most weather conditions [7], [17]-[19]. Common workflows compare pre-event and event composites, identify a decrease in VV backscatter, and exclude persistent water and steep terrain [1], [2]. A decrease can indicate newly exposed smooth water, but other processes can produce a similar signal. Land cover, radar shadow, double bounce, speckle, and acquisition timing all affect backscatter [2], [3]. A threshold mask is therefore an event-specific estimate that requires independent validation.

Where reference masks exist, datasets such as Sen1Floods11 permit conventional accuracy tests [8]. No comparable Kerala event mask was available for this study, so the SAR branch is reported as threshold-sensitive mapped area rather than a validated classification. Earlier Kerala studies demonstrate the value of event-scale mapping while noting that inundation extent captures only one dimension of impact [9].

B. Humanitarian Social Sensing

Crisis-informatics research classifies social media content by humanitarian purpose instead of relying only on keyword counts [4], [10]. HumAID supplies human labels for needs, infrastructure damage, displacement, rescue, and other disaster information. These categories capture consequences that are difficult to comprehend from overhead imagery. Their use still depends on successful geolocation and on who was able or willing to post [6]. Records can also be repeated or retrospective, and a place mentioned in the text describes the event rather than the author's location. Humanitarian labels do not remove these sampling and geolocation biases. District counts therefore represent observed reporting activity rather than population-normalized severity.

Social media data can help reconstruct flood conditions when satellite coverage is late or incomplete [11]. Exact coordinates are uncommon, copied reports are not independent, and multilingual place names are difficult to resolve. A district is

therefore assigned only when the text contains a recognized district or locality name. Other records remain unassigned.

C. Combining Remote and Human Observations

Remote and human observations have been combined in several ways. Geolocated news, photographs, and video have been used together with Landsat and elevation data [12]. Volunteered geographic information has been combined with EO-1 ALI imagery to build near-real-time flood-probability surfaces [13]. Satellite and Twitter frameworks have compared flood predictions with relevant text and images [14], [15]. Sentinel-1, Sentinel-2, social media hotspots, and exposed assets have also been placed on a common grid for response prioritization [16].

These studies show that remote and human observations can be complementary, but agreement between two noisy sources is not the same as ground truth. Disagreement may reflect acquisition time, semantic scope, location error, or participation bias [6], [16]. A fusion method should keep the source-specific evidence visible and expose how operational preferences affect the final order.

III. RESEARCH GAP, OBJECTIVES, AND CONTRIBUTIONS

A. Research Gap

Existing fusion research establishes that satellite and volunteered observations can improve situational awareness [12]-[16]. The remaining question is how to convert two incomplete district-level signals into an inspectable priority order when no authoritative district-priority labels are available. Existing studies commonly produce probability surfaces or common-grid overlays. A response list adds a further requirement. The effect of preference must remain visible, because no training target justifies a learned weight. Three issues are critical. Flood area and record counts have incompatible units. Agreement cannot be treated as independent validation. Threshold sensitivity within the SAR branch must also be separated from weight sensitivity across modalities. These choices can change absolute areas or rankings in different ways.

B. Objectives

This study evaluates whether rank-based fusion can preserve the distinction between mapped inundation and reported humanitarian impact. It tests how three SAR thresholds affect statewide area and district order, measures the association between balanced SAR area and impact-or-need record counts, and identifies which priority positions remain stable as the fusion weight changes. Agreement is interpreted as corroboration between inputs, while disagreement identifies cases that require further evidence.

C. Contributions

This work contributes a reproducible district-prioritization baseline for the 2018 Kerala flood. It keeps the Sentinel-1 and HumAID pipelines independent until district aggregation, brackets SAR results using sensitive, balanced, and conservative thresholds, restricts the social signal to district-resolved human labels for impact or need, and reports a complete fusion-weight sensitivity sweep. Agreement is treated as corroborating evidence, and disagreement is treated as a reason for closer review. Neither case is presented as validation.

IV. DATA AND METHODOLOGY

Figure 1 shows the two independent processing flows. The Sentinel-1 branch produces district flood-area ranks, and the HumAID branch produces district impact-or-need ranks. Fusion is applied only to these rank vectors.

A. Sentinel-1 Branch

This study uses Sentinel-1 Ground Range Detected (GRD) scenes processed in Google Earth Engine [17]-[19]. GRD is appropriate because the study relies on calibrated backscatter intensity and mapped area rather than phase or interferometric coherence. Single Look Complex data would add phase information without serving this change-detection method. Product characteristics are discussed in [7] and [17]. The scenes use VV polarization, descending passes, and Interferometric Wide mode. Kerala is partitioned into 81 tiles.

Reference and event composites use the periods of 20 July to 1 August and 15 to 31 August 2018. This intentional separation measures pre-flood to flood change.

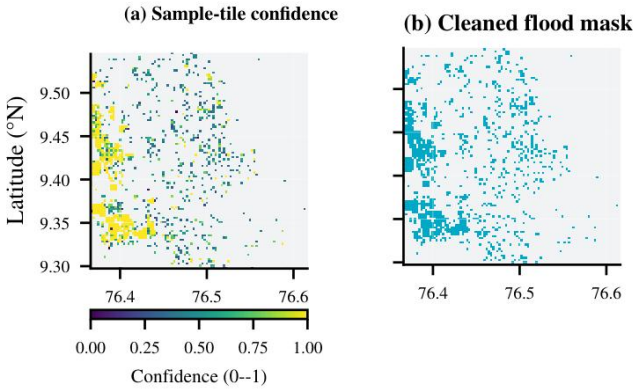
$$C_k(p) = \text{median}[q \in N(p)]\ \text{median}[t \in T_k]\ I_t(q).$$

Let It(p) denote VV backscatter in decibels at pixel p. For period k in {0,1}, Eq. (1) first takes the median across acquisition times Tk, then applies a focal median within neighbourhood N(p). The temporal median suppresses scene-specific noise. The spatial median reduces speckle-like variation without averaging across the full tile [1], [2]. C0 and C1 are the resulting reference and event composites.

Change is defined as $\Delta(p) = C1(p) - C0(p)$. Since the inputs are in decibels, this subtraction equals the logarithm of the intensity ratio, $10 \log10(\sigma0\text{event} / \sigma0\text{ref})$. It therefore retains a ratio interpretation while allowing thresholds to be stated in dB. Negative values indicate lower event-period backscatter, consistent with smooth open water but not conclusive on their own.

$$B\tau(p) = I[\Delta(p) < \tau]\ I[O(p) < 90]\ I[S(p) < 15°].$$

The thresholds −2.0, −2.5, and −3.0 dB define the sensitive, balanced, and conservative masks. Equation (2) also excludes pixels with JRC water occurrence at or above 90% or SRTM slope at or above 15° [20], [21].



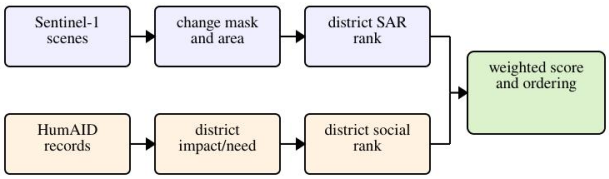


Fig. 1. Independent Sentinel-1 and HumAID processing flows join only at district-level weighted fusion.

Pixels are retained when the change passes τ, JRC water occurrence is below 90%, and SRTM slope is below 15°. The supporting layers have approximately 30 m native detail and use nearest-neighbour resampling. A nominal 10 m reduction scale does not increase their resolution. The −2.0, −2.5, and −3.0 dB thresholds are fixed sensitive, balanced, and conservative settings rather than values fitted to Kerala labels.

$$M\tau = (B\tau \oplus N_1) \ominus N_1.$$

Morphological closing is a post-threshold cleanup step. A one-pixel dilation followed by erosion connects adjacent detections and fills isolated one-pixel gaps. This reduces fragmentation in the binary mask without changing the threshold test. With N1 as the structuring element [22], Eq. (3) gives the cleaned mask.

$$K\tau(p) = M\tau(p) \cdot clip(-\Delta(p)/8, 0, 1).$$

In Eq. (4), $K\tau(p) \in [0,1]$ is the displayed change-strength score at pixel p. $M\tau(p)$ is the cleaned binary decision, and $\Delta(p)$ is the VV change in dB. The constant 8 dB sets the decrease at which the normalized magnitude reaches one. The operator $clip(x,0,1) = \min(1, \max(0,x))$ bounds its argument. Multiplication by $M\tau(p)$ sets rejected pixels to zero, so the visualization records change strength only inside the accepted mask.

TABLE I. PROCESSING CHOICES AND AVAILABLE RECORDS

Item	Value
Sentinel-1 selection	GRD, VV, descending, IW
Reference / event windows	20 Jul-1 Aug / 15-31 Aug 2018
Change thresholds	-2.0, -2.5, -3.0 dB
Water / terrain exclusions	occ. ≥ 90% / slope ≥ 15°
Focal median / area scale	30 m / 10 m
Dedup. HumAID records	7,984
District-resolved records	1,080
Impact/need records used	305

Fig. 2. Retained SAR products and their aggregation. (a) Change-strength score for a representative processing tile (Tile 016), displayed only for pixels accepted by the cleaned mask. (b) Balanced binary flood mask. (c) Balanced mapped area for all 81 tiles. (d) Statewide area under the three thresholds. Pre-event and event composites were not retained as image exports.

At a nominal 10 m reduction scale, tile area is

$$A_{r,\tau} = 10^{-6} \Sigma[p \in r]\, M\tau(p)\, a(p).$$

Here, A_r,τ is the mapped flood area in km² for tile r and threshold τ. $p \in r$ indexes pixels inside tile r. $M\tau(p)$ is the binary cleaned mask, and a(p) is the pixel area in m². The summation adds the accepted pixel areas, and the factor 10^{-6} converts m² to km². Tiles are linked to FAO GAUL level-2 districts [23]. Eighty tiles use the district containing the tile centroid, and one tile uses the district with the largest intersection. District area is $A[d,\tau] = \Sigma[r \in R[d]]\, A[r,\tau]$, where d indexes a district and R[d] is the set of tiles assigned to it. This whole-tile assignment is a boundary approximation rather than a pixel-district overlay.

B. Humanitarian Branch

The social data are the Kerala 2018 event records from HumAID [5]. After exact-text deduplication, 7,984 messages remain. District extraction searches for the 14 district names, selected locality aliases, hashtag and camel-case forms, and selected Malayalam names. A district is recovered from 1,080 messages (13.53%). Of the rest, 6,506 mention Kerala without a district, and 398 contain no recognized place. All later social results refer to this resolvable subset.

Five supplied labels are treated as evidence of local impact or need. These are requests or urgent needs (203 district-resolved records), infrastructure and utility damage (64), injured or dead people (22), displaced people and evacuations (16), and missing or found people (0). The combined count is 305. Rescue or donation activity, sympathy, caution, general

information, and non-humanitarian records are not used in the score. The available event records do not provide usable timestamps, so no claim is made about day-to-day alignment with acquisitions. Message text is not reproduced.

Let Hd denote the retained message count for district d. Names and aliases are canonicalized before counting. This avoids treating spelling variants as different districts, but it cannot recover reports that contain no recognizable locality.

C. Ranking and Evaluation

Because area and record counts have incompatible units, each is converted to a normalized within-event rank before fusion [24]. For D = 14 districts,

$$R^{dA}=[rank\uparrow(A[d,-2.5])-1]/(D-1),$$
$$R^{dH}=[rank\uparrow(H[d])-1]/(D-1).$$

In Eq. (6), RAd and RHd are the normalized satellite and humanitarian ranks for district d. Ad,−2.5 is the mapped area at the balanced threshold, and Hd is the retained impact-or-need record count. rank↑(x) returns the average ascending rank. A value of 1 marks the highest within-event position, and 0 marks the lowest. Ranking avoids treating one square kilometre as numerically comparable to one social media record, although it discards the distances between adjacent values. The combined score is

$$Q[d](w) = w\,R^{dA} + (1 - w)\,R^{dH},\ 0 \le w \le 1.$$

Here, Qd(w) is the fused score for district d. The weight w controls the satellite rank, and 1 − w controls the humanitarian rank. This study reports w = 0.5 as an equal-emphasis reference. A sweep from 0 to 1 in steps of 0.05 shows how priorities change as decision-makers place more emphasis on mapped inundation or reported impact.

SAR sensitivity is evaluated through statewide totals and pairwise district-rank correlations. Cross-modal association uses Spearman's ρ between balanced area and Hd. Its two-sided Monte Carlo probability is calculated from 100,000 random permutations.

$$p_{perm}=\{1+\Sigma[b=1...N\pi]I(|\rho(A,H\pi^b)|\ge|\rho obs|)\}/(N\pi+1).$$

In Eq. (8), p_{perm} is the estimated two-sided permutation probability, and Nπ is the number of permutations. The index b denotes one permutation, and π^b is the bth random reordering of district labels. A = (A[d,-2.5]) represents the D balanced district areas, and H = (H[d]) is the message-count vector. $H\pi^b$ is H reordered by π^b, and $\rho(A,H\pi^b)$ is the Spearman correlation after reordering. ρobs = ρ(A,H) is the observed correlation, |·| denotes absolute value, and I[·] counts permutations at least as extreme as the observation. The added ones in the numerator and denominator prevent a reported probability of zero [25]. A paired district bootstrap with 50,000 resamples supplies a percentile interval for ρ [26].

V. RESULTS

The analysis first evaluates SAR sensitivity. It then compares the two district patterns and examines how the fused order changes with weight.

A. What the SAR Branch Retains

Figure 2(a)-(b) shows a representative sample tile (Tile 016). Panel (a) displays the bounded change-strength score from Eq. (4) only where the cleaned balanced mask accepts a pixel. Panel (b) shows the corresponding binary flood mask. The archive does not contain the original reference and event composites, so they are not reconstructed.

The sample tile contributes 84.80 km² at the balanced setting and is the largest tile in Fig. 2(c). Statewide mapped areas are 344.74, 273.28, and 219.42 km² for the sensitive, balanced, and conservative settings. The sensitive estimate is 26.15% above the balanced estimate, while the conservative estimate is 19.71% below it. Estimated flood area therefore depends strongly on threshold choice.

District ordering is far more stable. Spearman correlations are 0.991 for sensitive versus balanced, 0.996 for balanced versus conservative, and 0.982 for sensitive versus conservative, each with $p < 10^{-9}$. The thresholds change the amount mapped more than the location of the largest concentrations.

Balanced district areas are led by Alappuzha (99.04 km²), Kottayam (55.24), Thrissur (42.24), Kannur (14.82), and Wayanad (11.38). Whole tiles are assigned to districts by the tile centroid. The large sample-tile contribution therefore also illustrates the boundary approximation introduced by that spatial partition.

B. The Two District Patterns Do Not Coincide

Balanced SAR area and impact-or-need message count have ρ = 0.279. The permutation probability is 0.330, and the district-bootstrap 95% interval is [−0.34, 0.77]. With only 14 districts, this interval is necessarily broad. More importantly, the observed data do not support a monotonic relationship strong enough to use one stream as a substitute for the other.

The scatterplot in Fig. 3(a) shows where the difference arises. Ernakulam has 114 retained messages, the largest social count, but stands eighth by balanced SAR area. Kottayam is second by area and eighth socially. Pathanamthitta is third socially and eleventh physically. Kannur presents the reverse case, ranking fourth by area while having no selected district-resolved impact-or-need record. These are not merely ranking anomalies. They are the cases in which a responder would want finer imagery, locality-level reports, or field evidence.

C. Priority Order Depends on the Weight

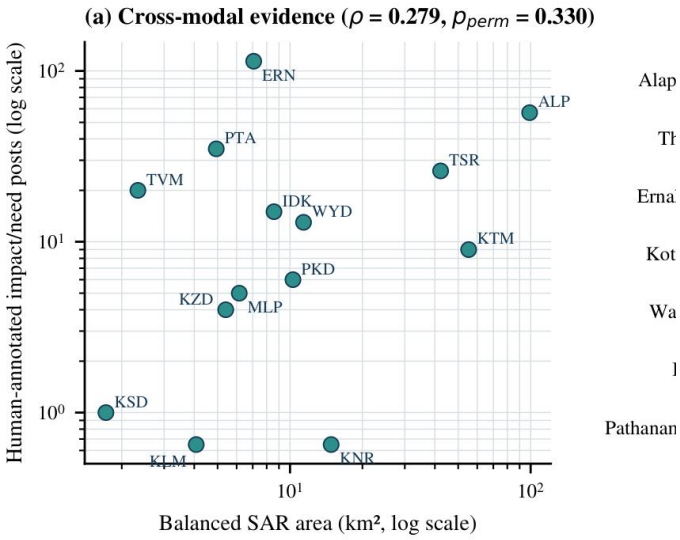


Fig. 3. District comparison and equal-weight fusion. (a) Balanced mapped area against retained impact/need records. KLM and KNR have zero selected records and are placed at 0.65 solely for log-scale display. (b) The dark and teal portions of each bar are the satellite and humanitarian rank contributions.

TABLE II. HIGHEST SCORES AT W = 0.5

District	A(km²)	H	r^A	r^H	Q
Alappuzha	99.04	57	1	2	0.962
Thrissur	42.24	26	3	4	0.808
Ernakulam	7.06	114	8	1	0.731
Kottayam	55.24	9	2	8	0.692
Wayanad	11.38	13	5	7	0.615
Idukki	8.58	15	7	6	0.577
Pathanamthitta	4.94	35	11	3	0.538

At w = 0.5, Alappuzha leads with Q = 0.962. Thrissur (0.808), Ernakulam (0.731), Kottayam (0.692), and Wayanad (0.615) complete the first five, followed by Idukki (0.577) and Pathanamthitta (0.538). Alappuzha combines the highest physical rank with the second-highest social rank. Ernakulam moves upward because its large social count offsets its eighth-place physical position, whereas Kottayam remains high for the opposite reason.

The order is not invariant to policy. At w = 0.25, Alappuzha, Ernakulam, Thrissur, Pathanamthitta, and Idukki form the first five. At w = 0.75, the order becomes Alappuzha, Thrissur, Kottayam, Wayanad, and Ernakulam. Figure 4 traces these changes. Alappuzha remains first for $w \geq 0.125$, and the lower positions reveal the trade-off rather than a numerical instability.

VI. DISCUSSION

Two forms of sensitivity emerge. Changing the SAR threshold substantially alters the estimated area but hardly changes the district ordering. Changing the fusion weight has the opposite practical effect, moving districts that are prominent in only one evidence stream. A logistics team concerned with widespread inundation may reasonably choose a larger w, while a team screening urgent reported consequences may choose a smaller one. The calculation makes that judgement visible rather than embedding it inside a trained model.

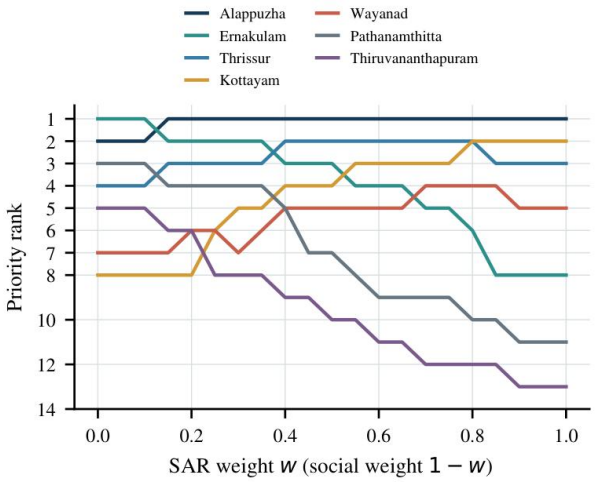


Fig. 4. District rank as emphasis shifts from humanitarian records to SAR area. Seven districts relevant to the leading positions are displayed.

Weak cross-modal association is not evidence that either dataset is useless. The two variables have different meanings. Open-water extent is shaped by radar physics and acquisition timing, while message counts are shaped by exposure, population, connectivity, language, and attention. Ernakulam's position, for example, invites investigation of urban impacts or local flooding not represented by a simple open-water rule. A high-area district with few reports may instead indicate sparse online coverage. Fusion is valuable here because it preserves these tensions rather than resolving them artificially.

Rank aggregation also has a cost. It prevents direct addition of incompatible units and limits the influence of extreme magnitudes, but it discards distance. A minor difference between two districts and a major difference can each cause the same one-rank change. Operational deployment should

compare Eq. (7) with value functions and weights elicited from disaster-management specialists.

No independent Kerala reference mask was available, so the SAR results are threshold-bracketed estimates rather than validated classifications. Stored raster exports include change, confidence, and masks, but not the reference and event composites, so the archived image chain is incomplete. Whole-tile district assignment introduces boundary error. Only 13.53% of messages resolve to a district, the gazetteer can miss uncommon spellings or localities, and the single-platform corpus is not population-normalized. Missing timestamps prevent acquisition-level temporal matching. Finally, no authoritative district-priority target exists against which Qd can be scored.

These gaps suggest several concrete next steps. Future work could intersect masks with administrative polygons at the pixel level, retain every intermediate raster, evaluate the method against independent flood labels, add rainfall, river-gauge, and field observations, improve multilingual locality resolution, and normalize online activity using population or platform baselines. More complex language or segmentation models would be useful only after suitable reference data make their improvement measurable.

VII. CONCLUSION

For the 2018 Kerala flood, satellite-derived area and district-resolved humanitarian messages produce related but distinctly different spatial patterns. The satellite district order stays stable across three change thresholds even though estimated extent varies by 125.33 km². Rank-based fusion makes their combination possible without mixing raw units, but the resulting list changes with the operational weight and is not a ground-truth allocation. Its defensible use is as a transparent screening device. Districts supported by both streams rise in priority, and districts where the streams diverge are made visible for further review.

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